

Waterfall Model

Overview

The **Waterfall Model** is one of the earliest and most traditional models in software engineering. It follows a *sequential design process*, where progress flows in one direction—downwards—through distinct phases. Each phase must be completed before the next begins, with little to no overlap between stages.

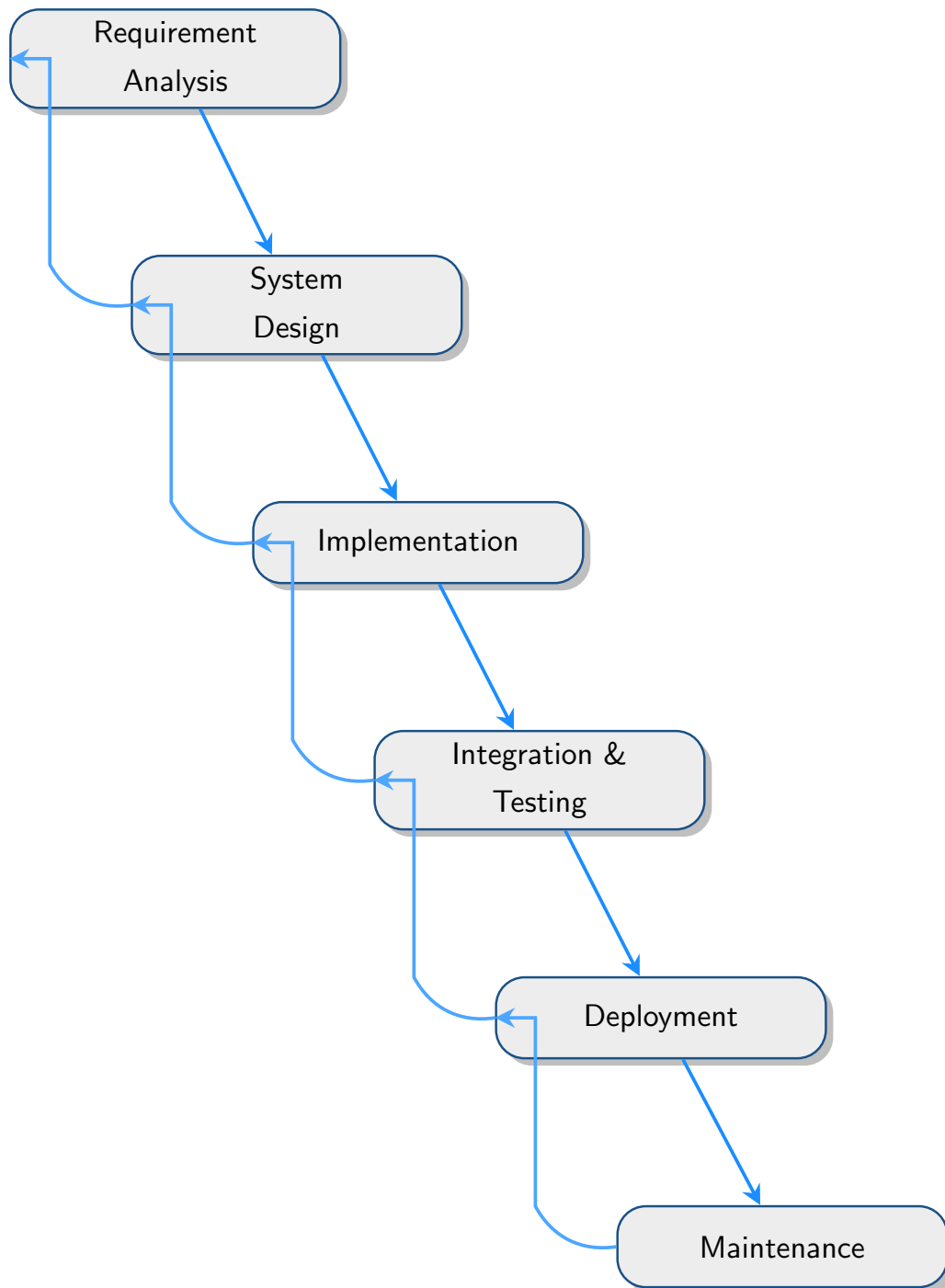
- It is a **plan-driven** process model.
- Each phase produces specific deliverables and documentation.
- Emphasizes disciplined development and clear milestones.

Phases of the Waterfall Model

The model consists of several well-defined stages, each with clear objectives:

1. **Requirement Analysis:** Gather and document customer requirements in detail.
2. **System Design:** Define system architecture, data flow, and overall structure.
3. **Implementation:** Develop source code based on design specifications.
4. **Integration & Testing:** Combine all modules and test the system for errors.
5. **Deployment:** Install the system for end-users and ensure proper operation.
6. **Maintenance:** Fix bugs, improve performance, and adapt to new requirements.

Waterfall Model Flow Diagram



Advantages

- Simple and easy to understand and manage.
- Well-suited for small, well-defined projects.

- Each phase has specific deliverables and review process.
- Documentation is maintained throughout development.

Disadvantages

- Difficult to accommodate changes once a phase is completed.
- Not suitable for complex, dynamic, or long-term projects.
- Late discovery of issues during testing phase.
- Customer feedback is received only at the end of the process.

When to Use the Waterfall Model

- When requirements are well-understood and fixed.
- When the technology and domain are well-known.
- For small projects with clear deliverables and minimal risk.